

CO1 Assessment 2

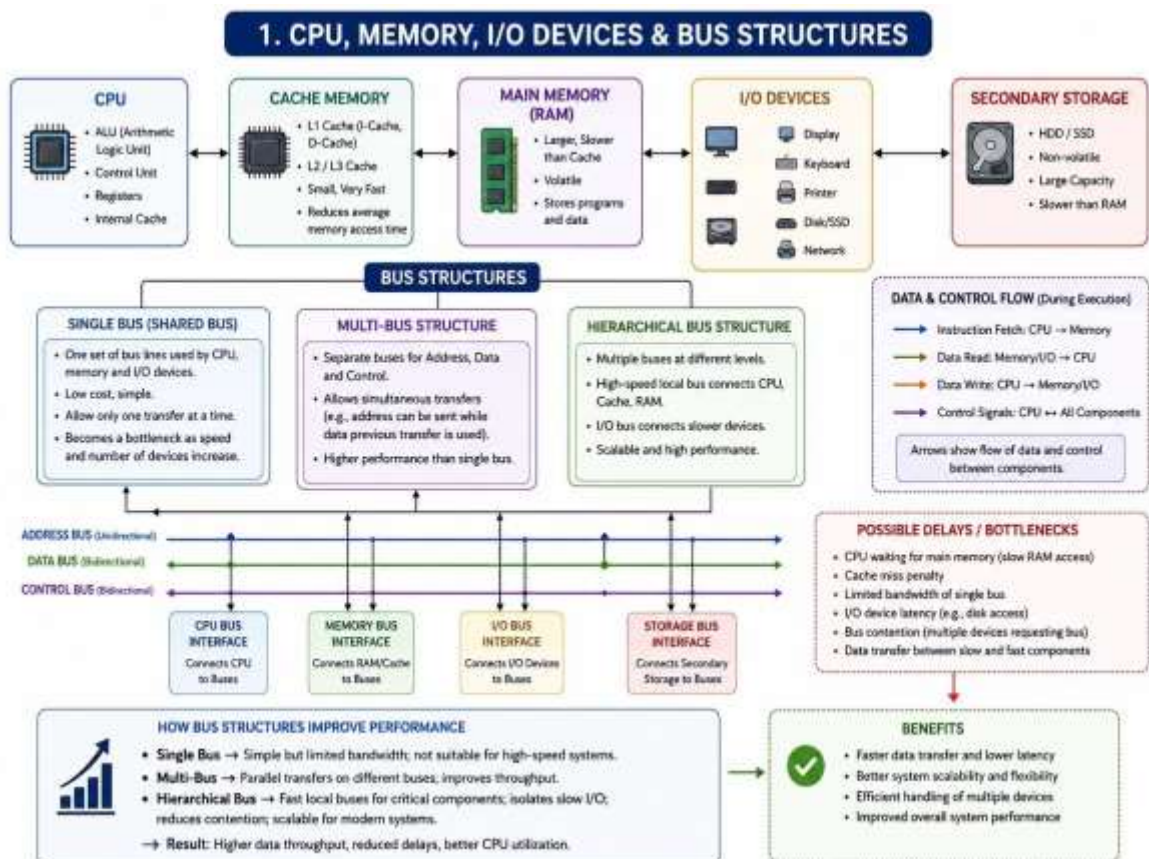
CONCEPT MAPPING

NAME : Shaswath A S

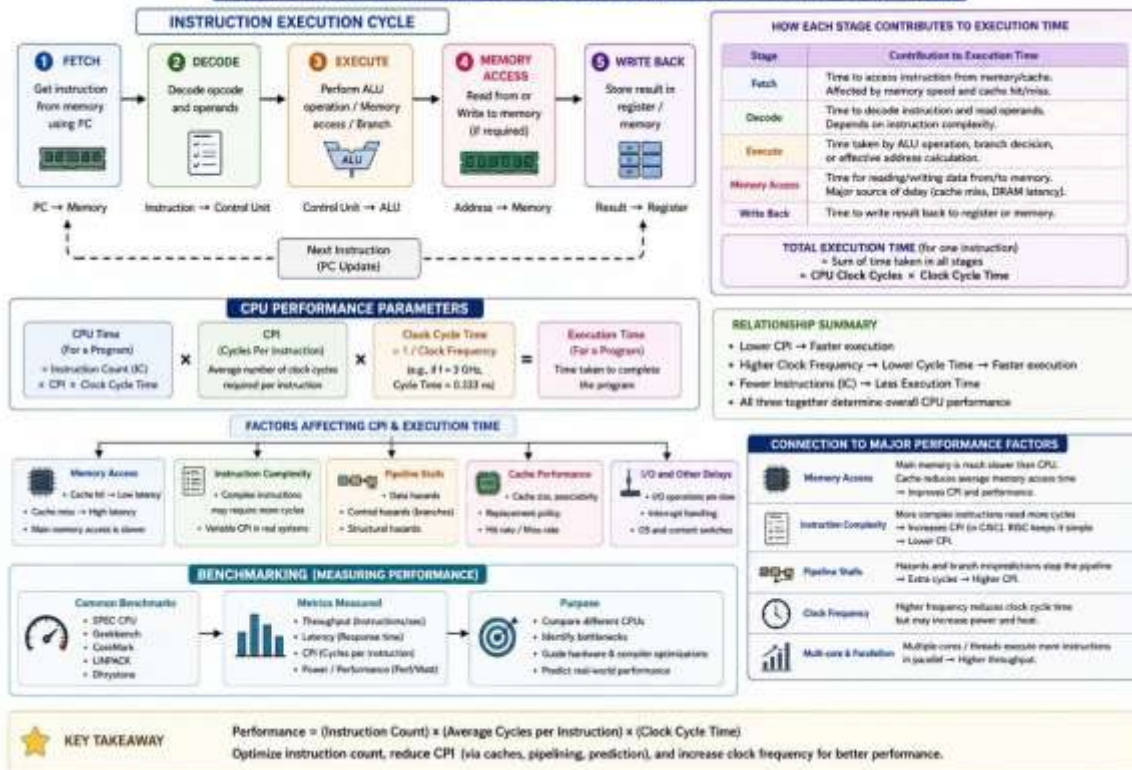
REG.NO : 192511195

CODE : CSA1210

COURSE : COMPUTER ARCHITECTURE



2. INSTRUCTION EXECUTION CYCLE & CPU PERFORMANCE



3. COMPARISON: 8085 vs 8086 ARCHITECTURES

ARCHITECTURAL OVERVIEW

8085 (8-bit Microprocessor)

- Introduced by Intel in 1976
- 8-bit data path
- Designed for embedded and control applications

8086 (16-bit Microprocessor)

- Introduced by Intel in 1978
- 16-bit data path
- Basis of x86 family
- Designed for high performance and multitasking

KEY DIFFERENCES AT A GLANCE

8085	8086
8-bit processor	16-bit processor
64 KB memory	1 MB memory
Fewer registers	More registers
No segmentation	Segmented memory
Simpler architecture	More complex architecture
Lower performance	Higher performance

FEATURE COMPARISON

FEATURE	8085	8086
Launch Year	1976	1978
Architecture	8-bit	16-bit
Data Bus Width	8 bits	16 bits
Address Bus Width	16 bits (64 KB memory)	20 bits (1 MB memory)
Registers	6 General Purpose (B, C, D, E, H, L) - Accumulator (A) - Flag Register	6 General Purpose (AX, BX, CX, DX, SP, BP, SI, DI) - Segment Registers (CS, DS, ES, SS) - IP (Instruction Pointer) & Flag Register
Register Size	8 bits	16 bits
Addressing Modes	5 Modes: Immediate, Register, Direct, Register Indirect, Implied	8 Modes: Immediate, Register, Direct, Register Indirect, Based, Indexed, Based Indexed, Relative, Direct/Displacement
Memory Segmentation	No	Yes (CS, DS, ES, SS) Supports 1 MB memory space
Instruction Set	74 Instructions	~100+ Instructions
Multitasking	Limited	Better (Segmented memory helps multitasking)
Pipeline	No	Basic BU (Bus Interface Unit) and EU (Execution Unit) allow overlap
I/O Addressing	8-bit I/O address (256 ports)	16-bit I/O address (64K ports)
Performance	Lower (8-bit)	Higher (16-bit, more registers, better memory handling)
Typical Applications	Small systems, Embedded, Control, Educational	Operating systems, Computers, Advanced applications

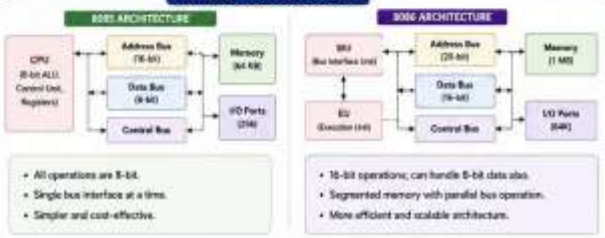
HOW FEATURES INFLUENCE PERFORMANCE

- Data Bus Width**
8086 transfers 16 bits at a time → Higher data throughput than 8085 (8 bits).
- Address Bus Width**
8086 can address 1 MB memory vs 64 KB in 8085 → Can handle larger programs.
- Registers**
8086 has more and larger registers → Fewer memory accesses → Faster execution.
- Memory Segmentation (8086)**
Allows code, data, stack in different segments → Better organization and multitasking.
- Addressing Modes**
8086 has more modes → Greater flexibility and efficient memory access.
- Pipeline (8086)**
BU and EU work in parallel → Faster instruction fetch and execution.

IMPACT ON EXECUTION & EFFICIENCY

- 8086 can execute more complex programs.
- Better memory management and multitasking.
- Higher throughput and faster real-time performance.
- 8085 is simpler, cost-effective, and suitable for small, dedicated tasks.

ARCHITECTURAL COMPONENTS



OVERALL SUMMARY

- 8086 is an 8-bit microprocessor suitable for simple, low-cost applications with limited memory needs.
- 8086 is a 16-bit microprocessor with advanced features like segmentation, more registers, and better performance.
- The 8086 architecture laid the foundation for modern x86 processors used in PCs and servers.

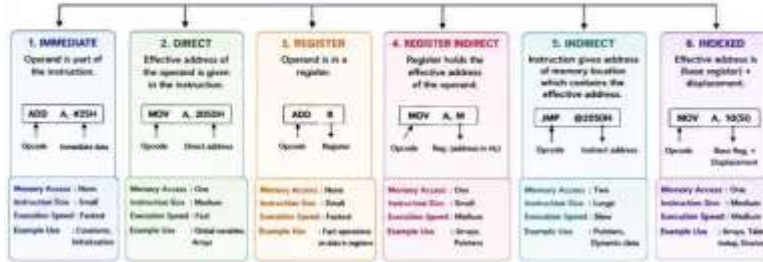
CONCLUSION

- 8086 is a more powerful and versatile architecture compared to 8085, offering higher performance, larger memory space, and better efficiency for complex and multitasking applications.

4. ADDRESSING MODES & INSTRUCTION TYPES

ADDRESSING MODES

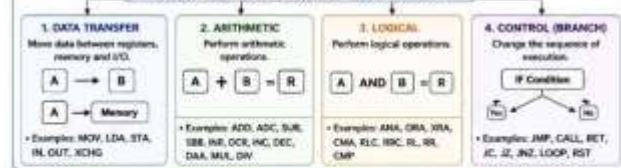
Addressing modes specify how the operand is located.



IMPACT SUMMARY				
Addressing Mode	Memory Accesses	Instruction Size	Execution Speed	When to Use
Immediate	None	Small	Fastest	Constants, Initialization
Direct	One	Medium	Fast	Global variables, Static data
Register	None	Small	Fastest	Fast operations
Register Indirect	One	Small	Medium	Arrays, Pointers
Indirect	Two	Large	Slow	Pointers, Dynamic data, Complex structures
Indexed	One	Medium	Medium	Arrays, Table lookup, Structures

INSTRUCTION TYPES

Instructions are classified based on the operation they perform.



REAL-WORLD SCENARIOS

Scenario	Preferred Addressing Mode(s)	Reason
Initializing a variable with a constant	Immediate	No memory access, fastest and smallest.
Accessing a global variable	Direct	Address is fixed and known.
Performing calculations in registers	Register	No memory access, maximum speed.
Traversing an array	Register Indirect / Indexed	Pointer in register or base+offset is efficient.
Following a pointer to a structure	Indirect	Address of address is needed.
Table lookup with offset	Indexed	Base address + offset is ideal.

HOW ADDRESSING MODES AFFECT PERFORMANCE



CHOOSING THE RIGHT ADDRESSING MODE



KEY TAKEAWAY

- Addressing mode determines how the operand is located. It directly affects memory access time, instruction size and execution speed.
- Choosing the right mode improves performance and reduces memory usage.
- Different modes are suitable for different programming scenarios.

5. NUMBER SYSTEMS, INSTRUCTION REPRESENTATION, MEMORY & DEBUGGING

